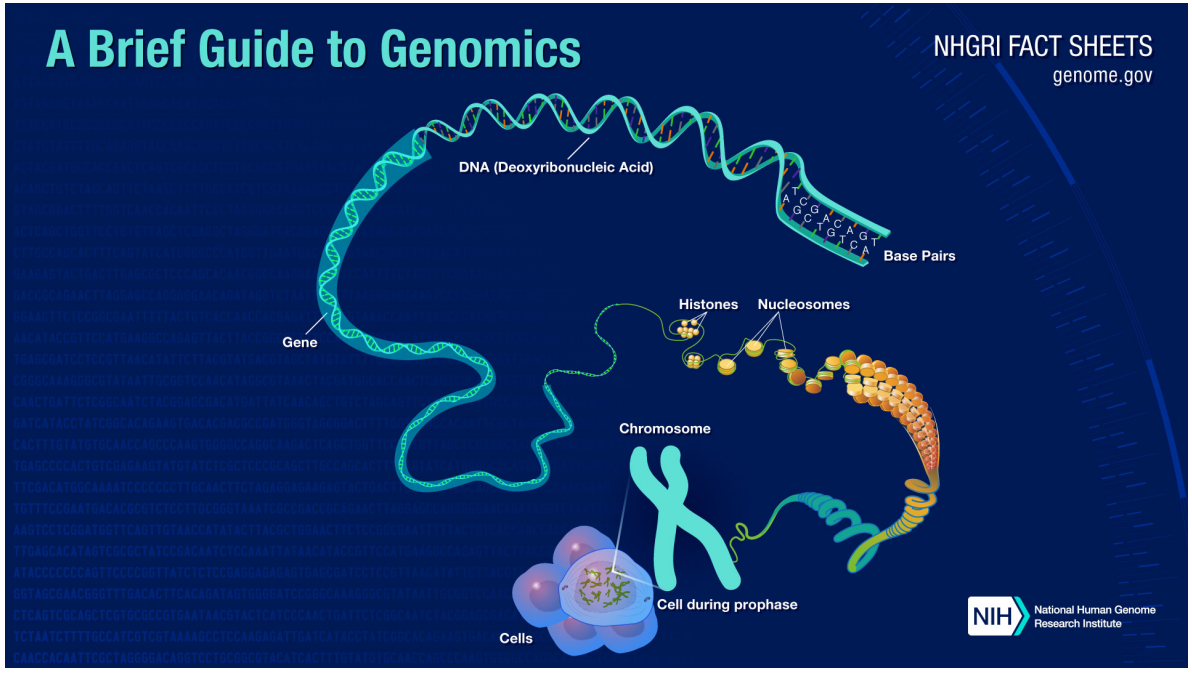
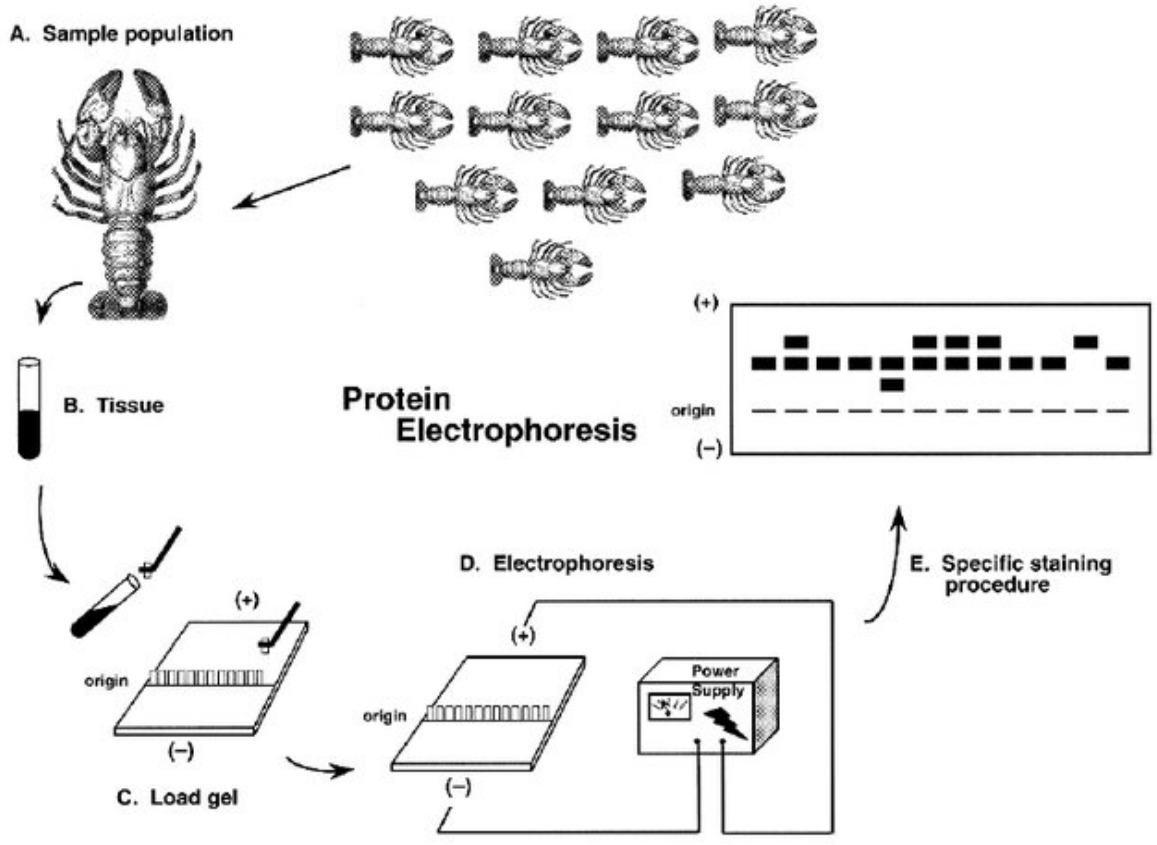
Assaying genetic diversity



- two copies of the nuclear genome found in most cells
- 100-1000 copies of mitochondrial genome found in most cells
- 3-275 copies of chloroplast genome per plastid (photosynthetic organelles considered endosymbiotic cyanobacteria)

(https://www.genome.gov/about-genomics/fact-sheets/A-Brief-Guide-to-Genomics).

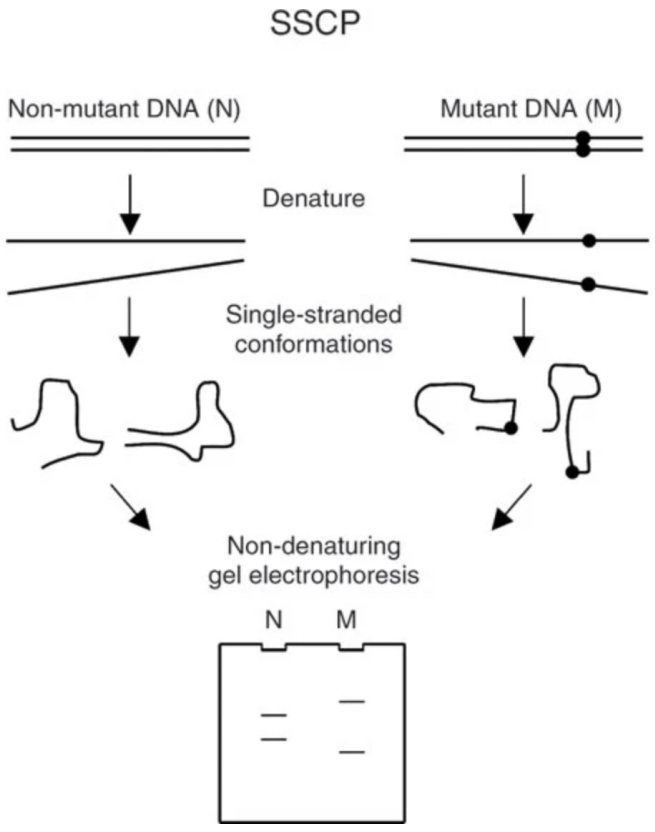
Technique: Protein electrophoresis



- alleles: pattern on gel lane
- by definition, proteins are **functional** (non-neutral, i.e. under selection)

(http://dx.doi.org/10.1038/238113b0).

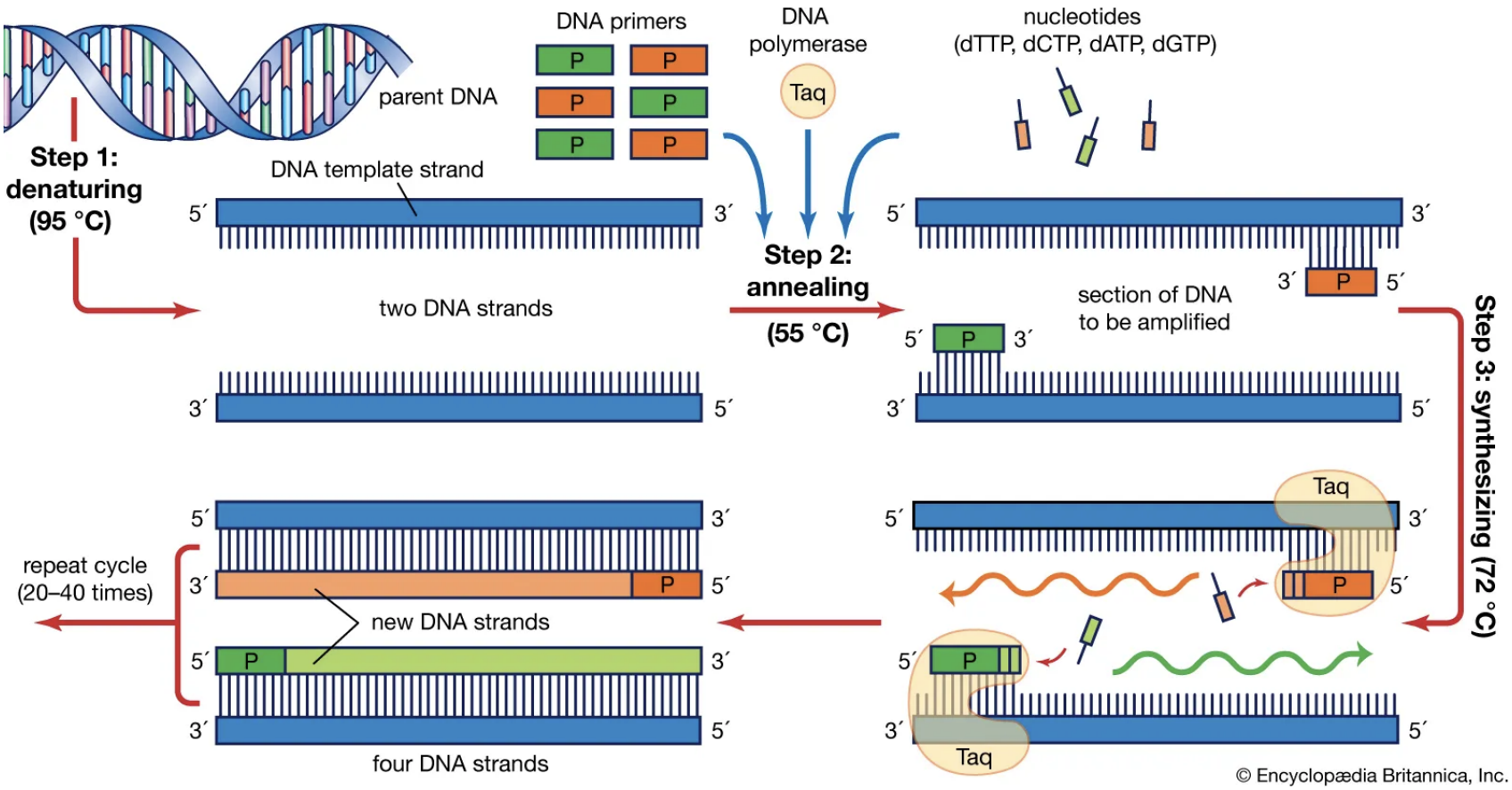
Technique: SSCP



- alleles: pattern on gel lane
- caused by differences in "single-strand conformation", i.e. how DNA bends differently with mutations

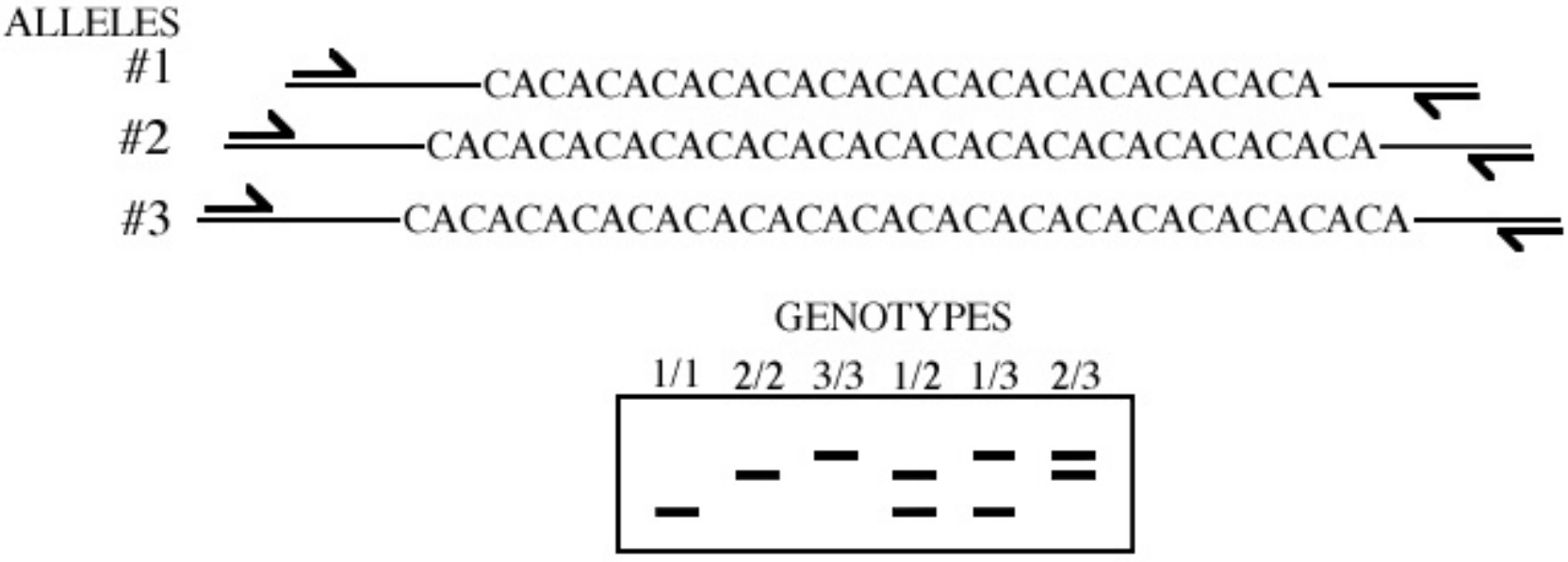
(https://doi.org/10.1038/nprot.2006.485).

Technology: PCR



(https://www.britannica.com/science/polymerase-chain-reaction).

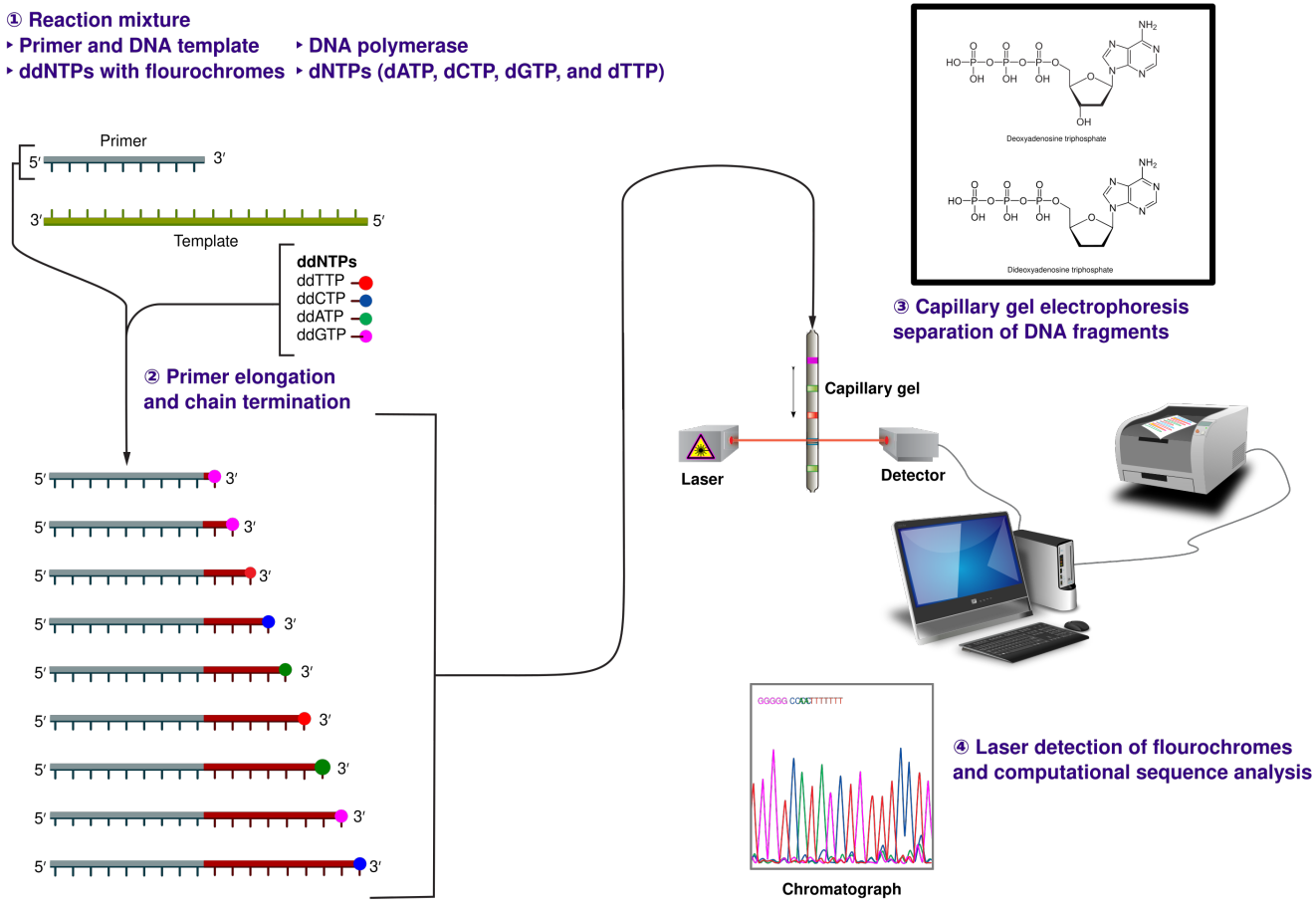
Technique: Microsatellites



- alleles: pattern on gel lane.
- almost always **neutral**.

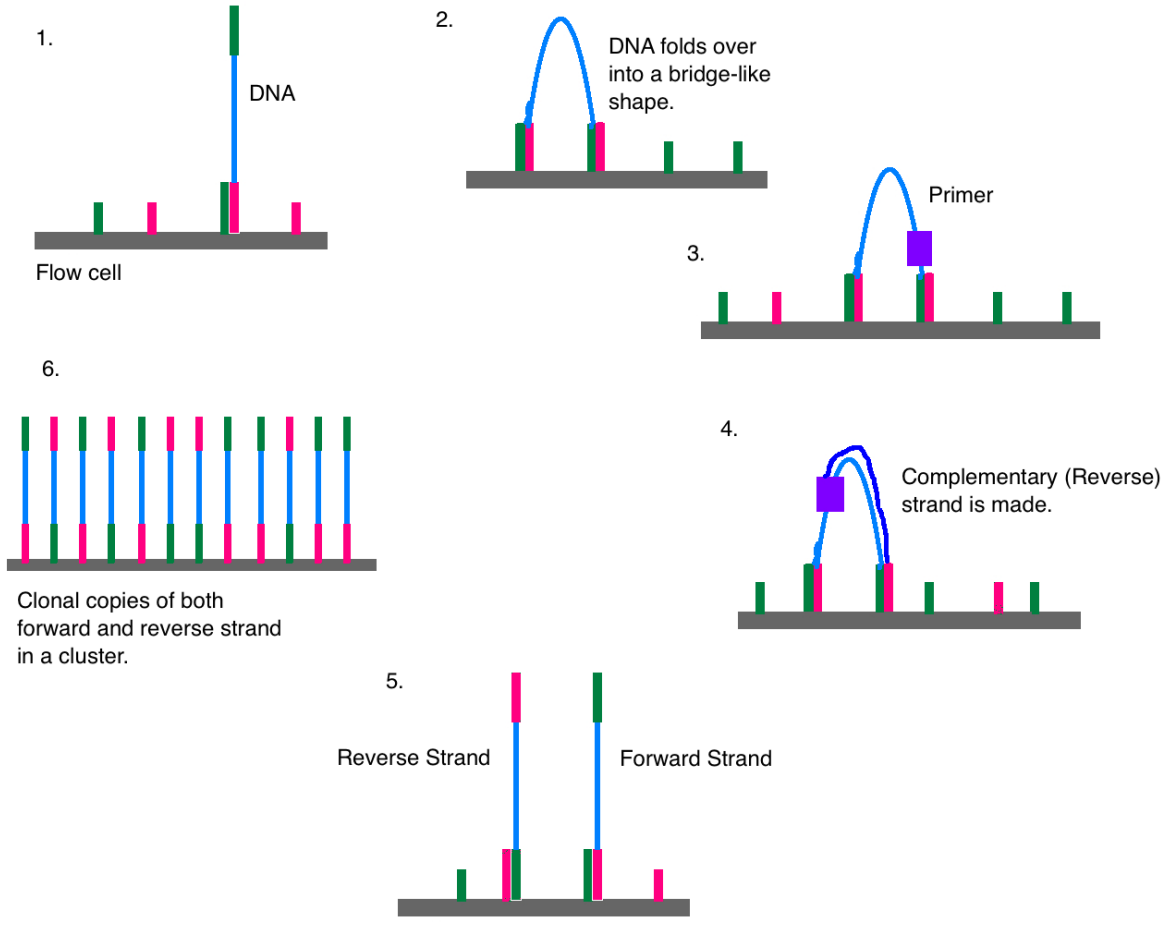
(https://www.informatics.jax.org/silver/figures/figure8-10.shtml).

Technology: Sanger sequencing



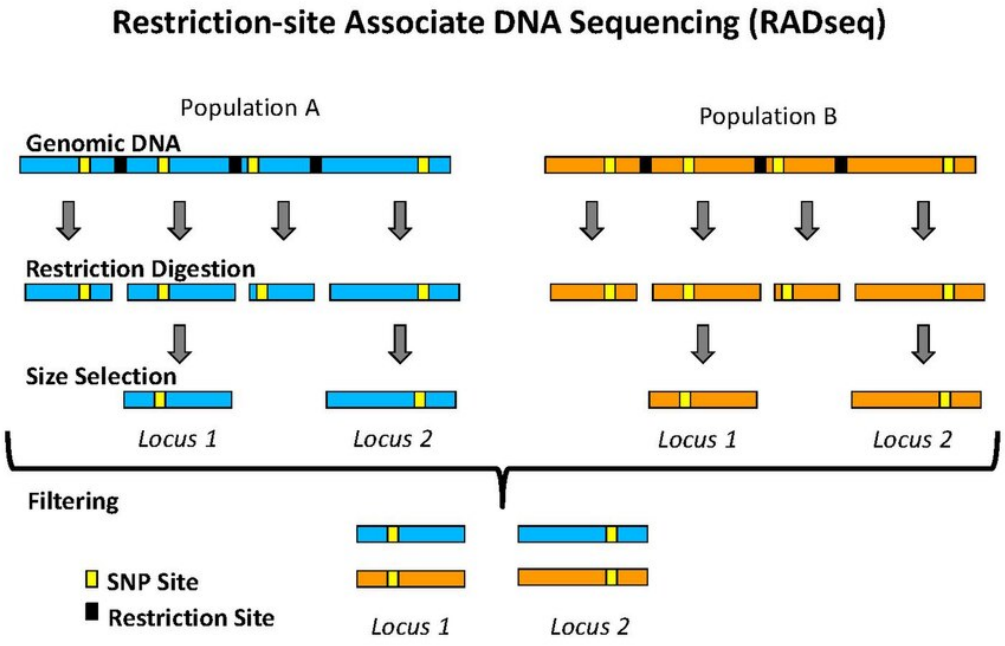
- combined with PCR, alleles are typically SNPs or haplotypes. (https://en.wikipedia.org/wiki/Sanger_sequencing).

Technology: Illumina sequencing



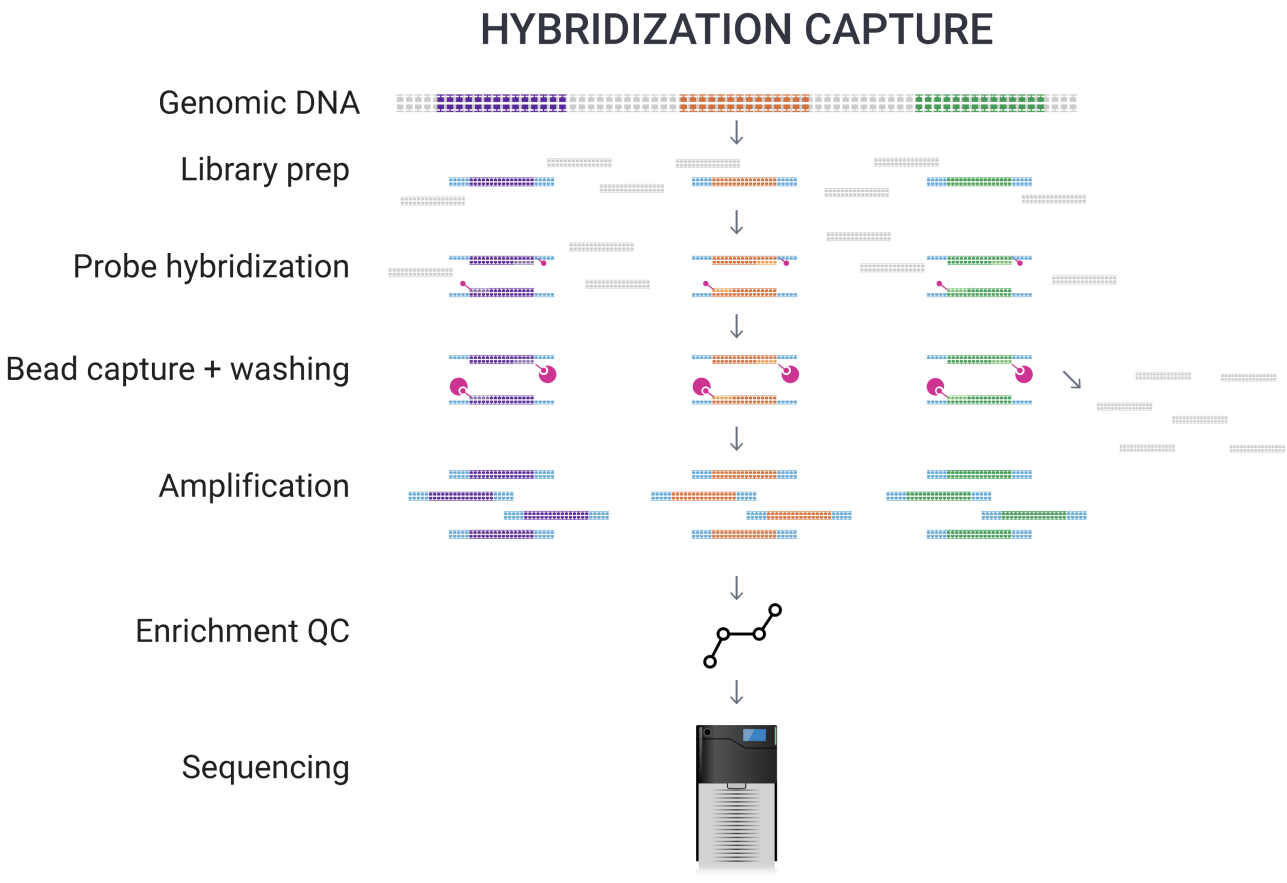
(https://en.wikipedia.org/wiki/Illumina_dye_sequencing).
(https://youtu.be/fCd6B5HRaZ8).

Technique: RADseq



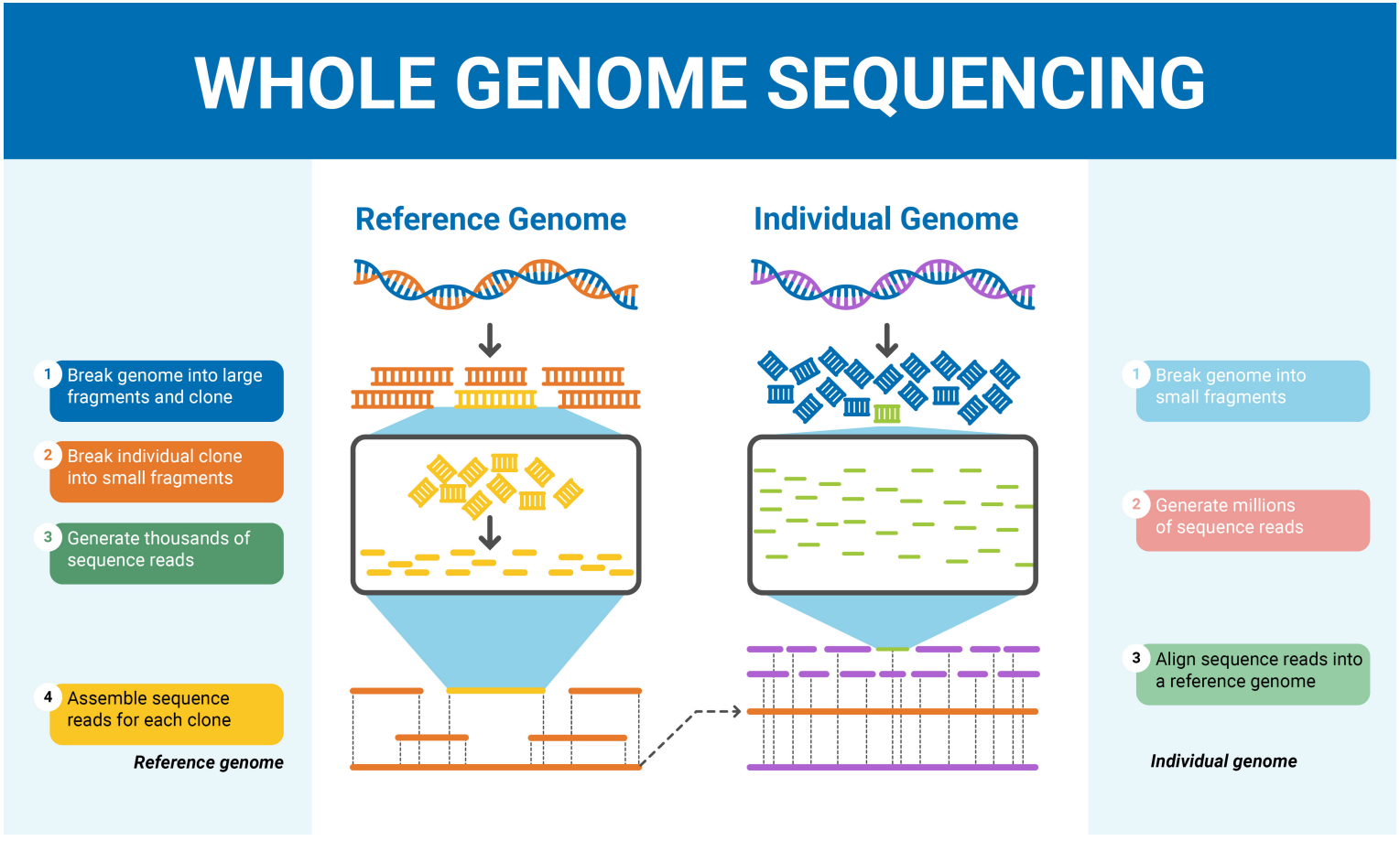
- alleles: SNPs (neutral or functional) (https://en.wikipedia.org/wiki/Restriction_site_associated_DNA_markers)

Technique: Sequence capture



- alleles: SNPs (neutral or functional). (https://www.pacb.com/products-and-services/applications/targeted-sequencing/).

Whole genome (re)sequencing

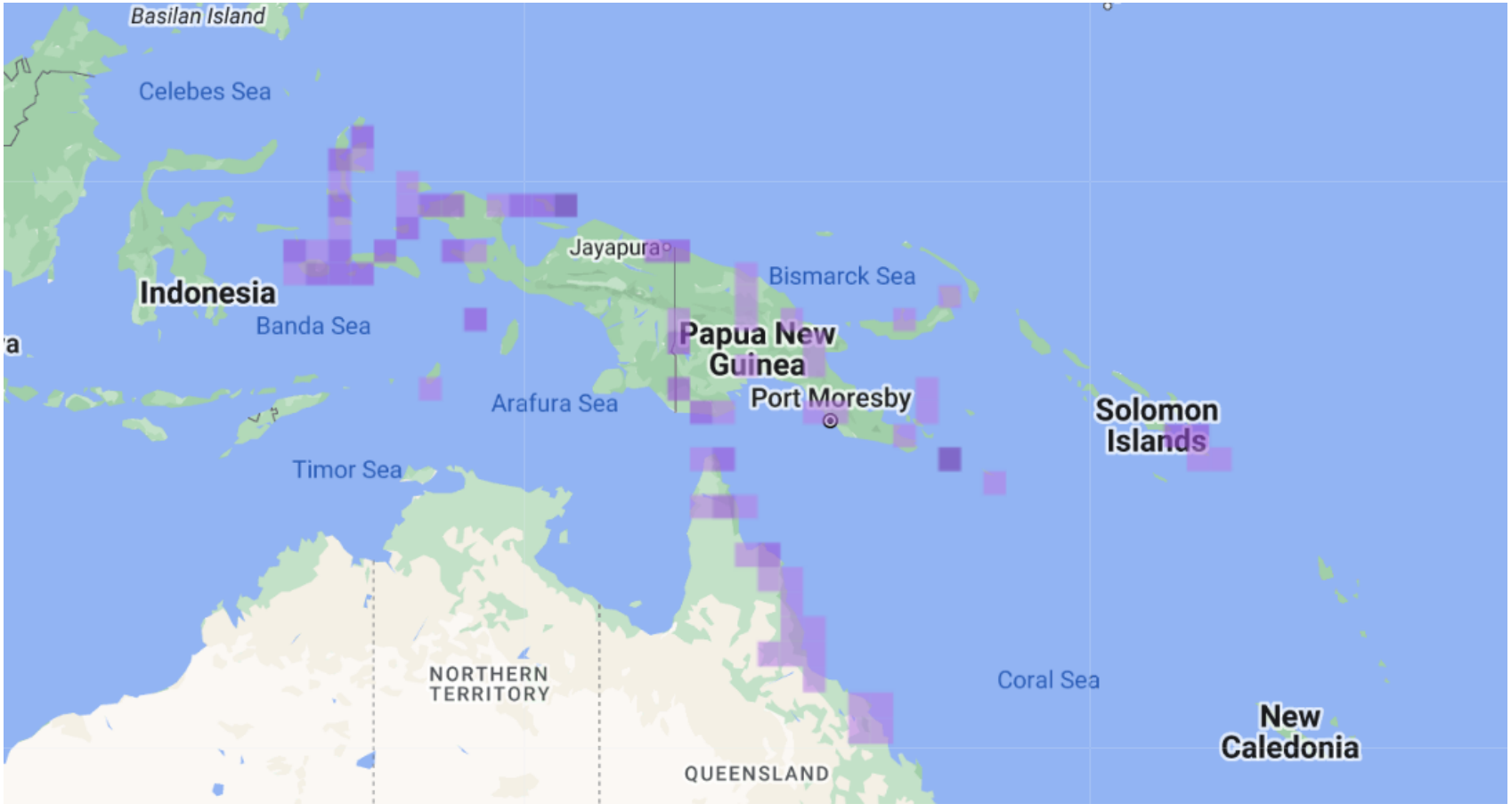


- alleles: SNPs, haplotypes, structural variants, copy number variants, etc.

Metallic Starling (*Aplonis metallica*)



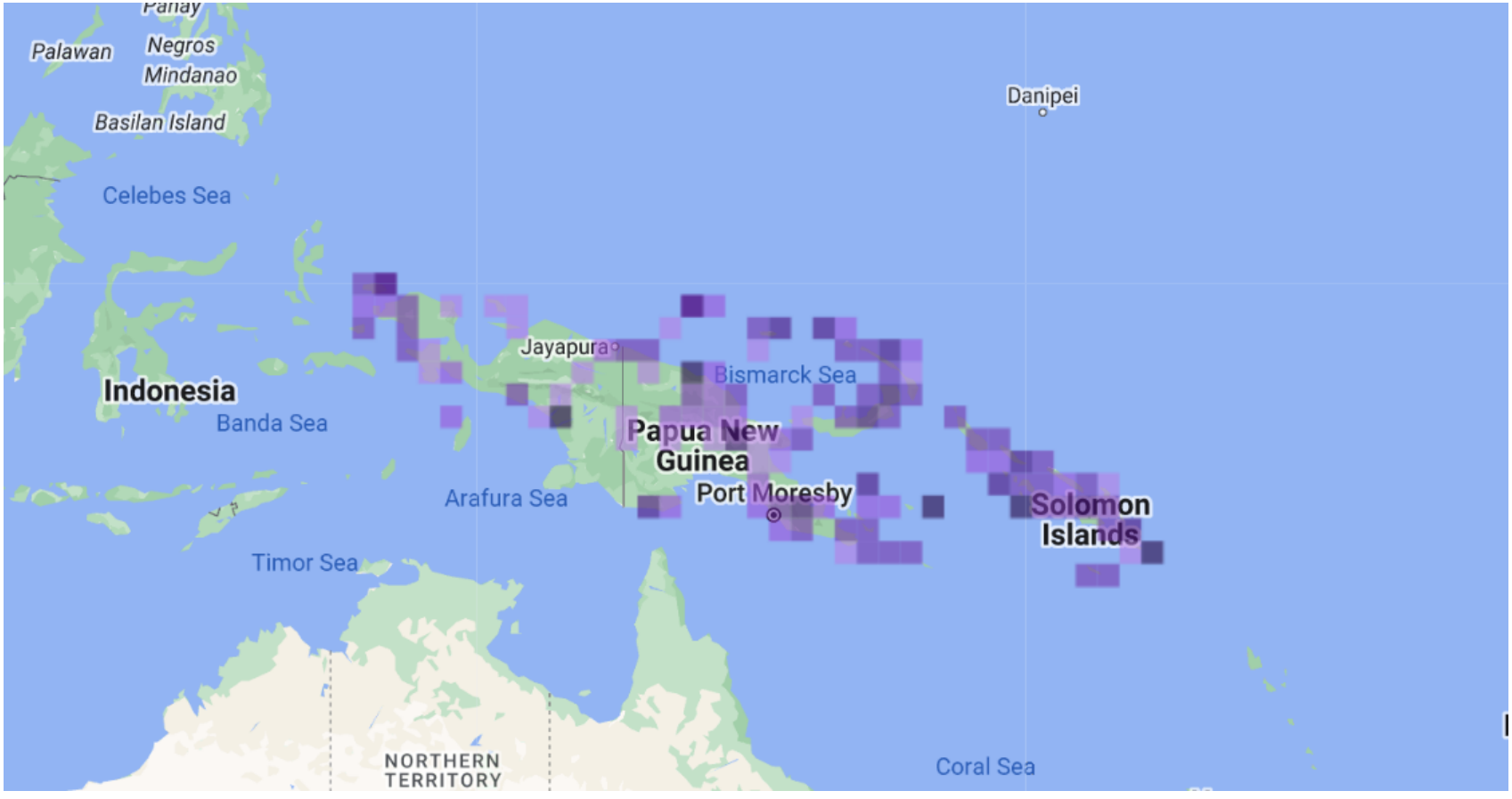
Metallic Starling (*Aplonis metallica*)



Singing Starling (*Aplonis cantoroides*)



Singing Starling (*Aplonis cantoroides*)



Group Questions

1) Why did the authors apply a chi-squared test, and what did they learn from it?
2) Why does the proportion of polymorphic proteins differ from the proportion of polymorphic loci?
3) What is the observed heterozygosity of *A. metallica* at LDH-H considering only samples from Lae, Madang, and Wewak?
4) Develop a hypothesis for why there is more polymorphism in *A. metallica* based on information in the reading (or other sources)
5) What is one takeaway from Figure 1?